

# Lecture 10: Clustering

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10 ECTS Credits

Department of Information and Communication Technology

Faculty of Engineering and Science

# Recap

- Dimensionality Reduction
- Feature Selection
  - Exhaustive Search
  - Univariate Greedy Search
  - Multivariate Greedy Search
    - Sequential Forward Search
    - Sequential Backward Search
    - Generalized Sequential Search
    - Plus- $l$  Minus- $r$  Search
    - Floating search
- Principle Component Analysis
  - PCA Steps
  - PCA Intuition
  - PCA Issues



# Clustering

- Unlabeled data is available: expensive labeling, reliable labeling not available, or data comes from a single group.
- Domain of **unsupervised learning**.
- Understanding the structure of underlying data distribution in the original feature space.
- Used to find **clusters** in data and build hierarchical data representations.
- If label is available, clustering can be used for detecting unknown classes.
- We are going to look at:
  - $K$ -means clustering,
  - Hierarchical clustering, and
  - Self-organizing maps (SOM)

# K-Means Clustering

- Given data  $S_n = \{X_1, \dots, X_n\}$ ,  $k$ -means algorithm finds  $K$  cluster centers  $\mu_1, \dots, \mu_K$  and for each point  $X_i$  find an assignment to one of  $K$  clusters.
- Cluster assignment is done by using vectors  $r_1, \dots, r_n$ .
- Each vector  $r_i$  is a vector of size  $K$  using one-hot encoding scheme.
$$r_i = (0, \dots, 1, \dots, 0)^T, \quad \forall i = 1, \dots, n$$
- Here  $r_i(k) = 1$ , iff.  $X_i$  belongs to cluster  $k$ ,  $\forall k = 1, \dots, K$ .
- Example:
  - Suppose  $K = 3$  and  $n = 4$ .
  - We may have  $r_1 = (0,1,0)$ ,  $r_2 = (0,0,1)$ ,  $r_3 = (1,0,0)$ , and  $r_4 = (0,1,0)$ .
  - Here  $X_1, X_4$  gets assigned to cluster 2,  $X_2$  is assigned cluster 3 and  $X_3$  is assigned cluster 1.

# K-Means Clustering

- The algorithm finds the vectors  $\{\boldsymbol{\mu}_k\}_{k=1}^K$  and  $\{\mathbf{r}_i\}_{i=1}^n$  that minimizes a score based on the normalized sum of distances of all points to their corresponding centers:

$$J = \frac{1}{n} \sum_{i=1}^n \sum_{k=1}^K \mathbf{r}_i(k) \|\mathbf{X}_i - \boldsymbol{\mu}_k\|^2$$

- The solution is obtained iteratively with two optimizations at each step:
  - With  $\{\boldsymbol{\mu}_k\}_{k=1}^K$  fixed,  $\{\mathbf{r}_i\}_{i=1}^n$  values that minimize  $J$  are found by inspection:

$$\mathbf{r}_i(k) = \begin{cases} 1, & \text{if } k = \arg \min_{j=1, \dots, K} \|\mathbf{X}_i - \boldsymbol{\mu}_j\|^2, \\ 0, & \text{otherwise} \end{cases}, \quad \forall i = 1, \dots, n$$

- With  $\{\mathbf{r}_i\}_{i=1}^n$  fixed,  $\{\boldsymbol{\mu}_k\}_{k=1}^K$  values that minimize  $J$  are found by differentiation:

$$\frac{\partial J}{\partial \boldsymbol{\mu}_k} = 2 \sum_{i=1}^n \mathbf{r}_i(k) \|\mathbf{X}_i - \boldsymbol{\mu}_k\|^2 = 0, \text{ which gives } \boldsymbol{\mu}_k = \frac{\sum_{i=1}^n \mathbf{r}_i(k) \mathbf{X}_i}{\sum_{i=1}^n \mathbf{r}_i^k}, \quad \forall k = 1, \dots, K$$

# K-Means Clustering

- In other words,
  - E-Step: In step 1, each point is assigned to the closest mean,
  - M-Step: In step 2,  $\mu_k$  is assigned the mean value of all the training points assigned to cluster  $k$  in step 1.

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**Algorithm 1** K-Means.

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1: Initialize  $K$ ,  $\tau > 0$  and  $\{\mu_k^{(0)}\}_{k=1}^K$ .

2: **repeat**

3:     E-Step: Update cluster assignments:

$$\mathbf{r}_i^{(m+1)}(k) = \begin{cases} 1, & \text{if } k = \arg \min_{j=1, \dots, K} \|\mathbf{X}_i - \mu_j^{(m)}\|^2, \\ 0 & \text{otherwise, } i = 1, \dots, n. \end{cases}$$

4:     M-Step: Update cluster centers:

$$\mu_k^{(m+1)} = \frac{\sum_{i=1}^n \mathbf{r}_i^{(m+1)}(k) \mathbf{X}_i}{\sum_{i=1}^n \mathbf{r}_i^{(m+1)}(k)}, \quad k = 1, \dots, K.$$

5:     Calculate score:

$$J^{(m+1)} = \frac{1}{n} \sum_{i=1}^n \sum_{k=1}^K \mathbf{r}_i^{(m+1)}(k) \|\mathbf{X}_i - \mu_k^{(m+1)}\|^2.$$

6: **until**  $|J^{(m+1)} - J^{(m)}| < \tau$ .

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# $K$ -Means Clustering

- One of the issues with  $K$ -means algorithm is the presence of multiple minima in the score function  $J$ .
- To find a good solution
  - the algorithm is run multiple times with random initialization of the centers,
  - Select solution with minimum score.
- This is known as **random restart method**.
- The choice of  $K$  is a difficult problem as large  $K$  compared to  $n$  creates overfitting.
- In practice,  $K$  is selected as that corresponding to the first occurrence of a sharp drop followed by stable values in a plot of  $J$  against  $K$ .

# Hierarchical Clustering

- $K$ -means algorithm produces a fixed assignment of data points to clusters.
- In hierarchical clustering, different clustering results are obtained by an iterative process of cluster creation.
- The algorithm could be:
  - **Agglomerative** (bottom-up): starting with each point in separate clusters and iteratively merge clusters.
  - **Divisive** (top-down): start with all the data in a single cluster and iteratively split clusters.

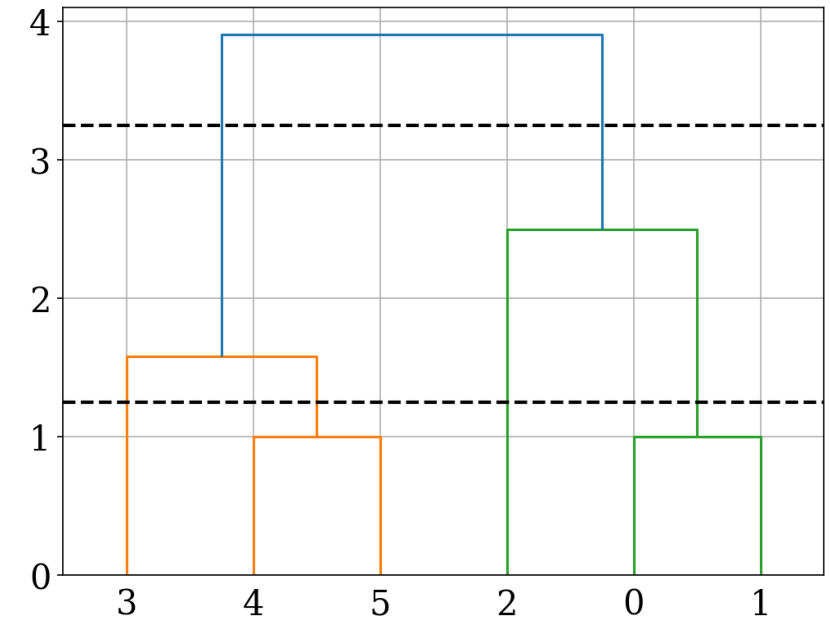


# Hierarchical Clustering

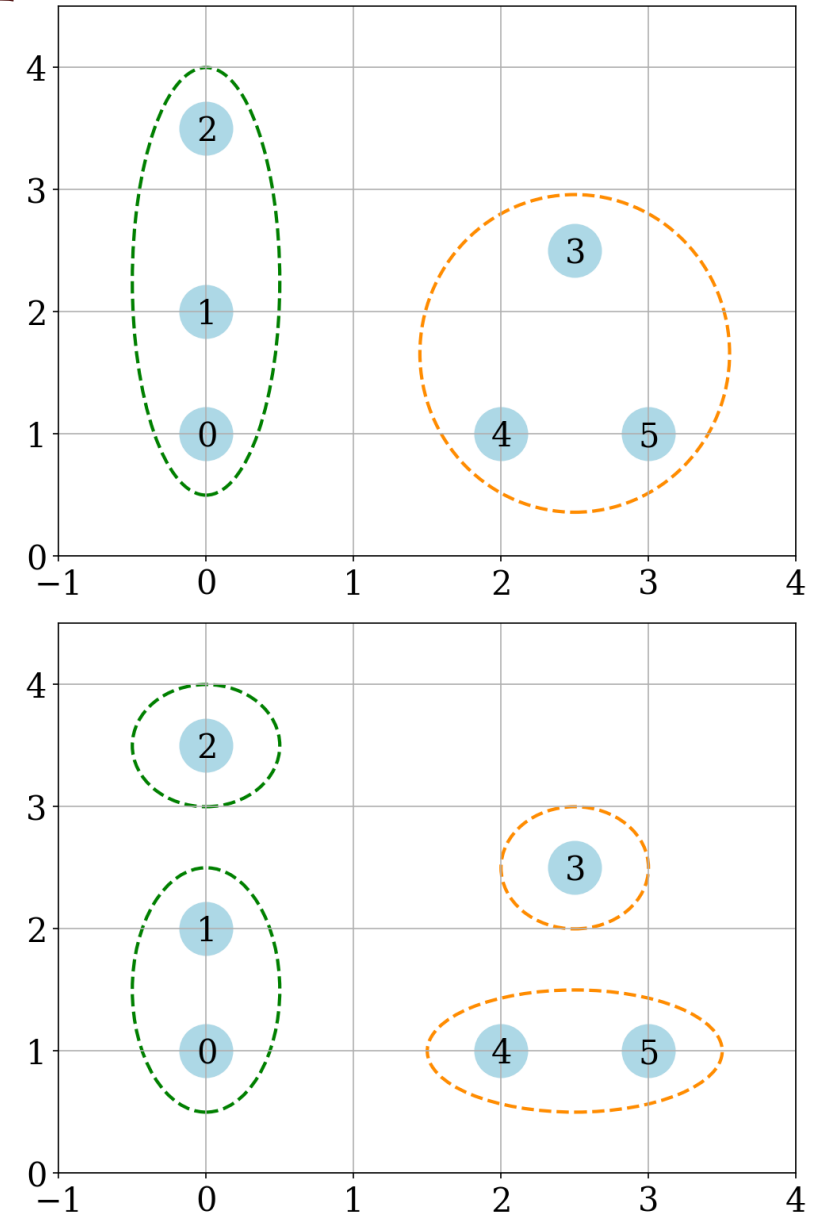
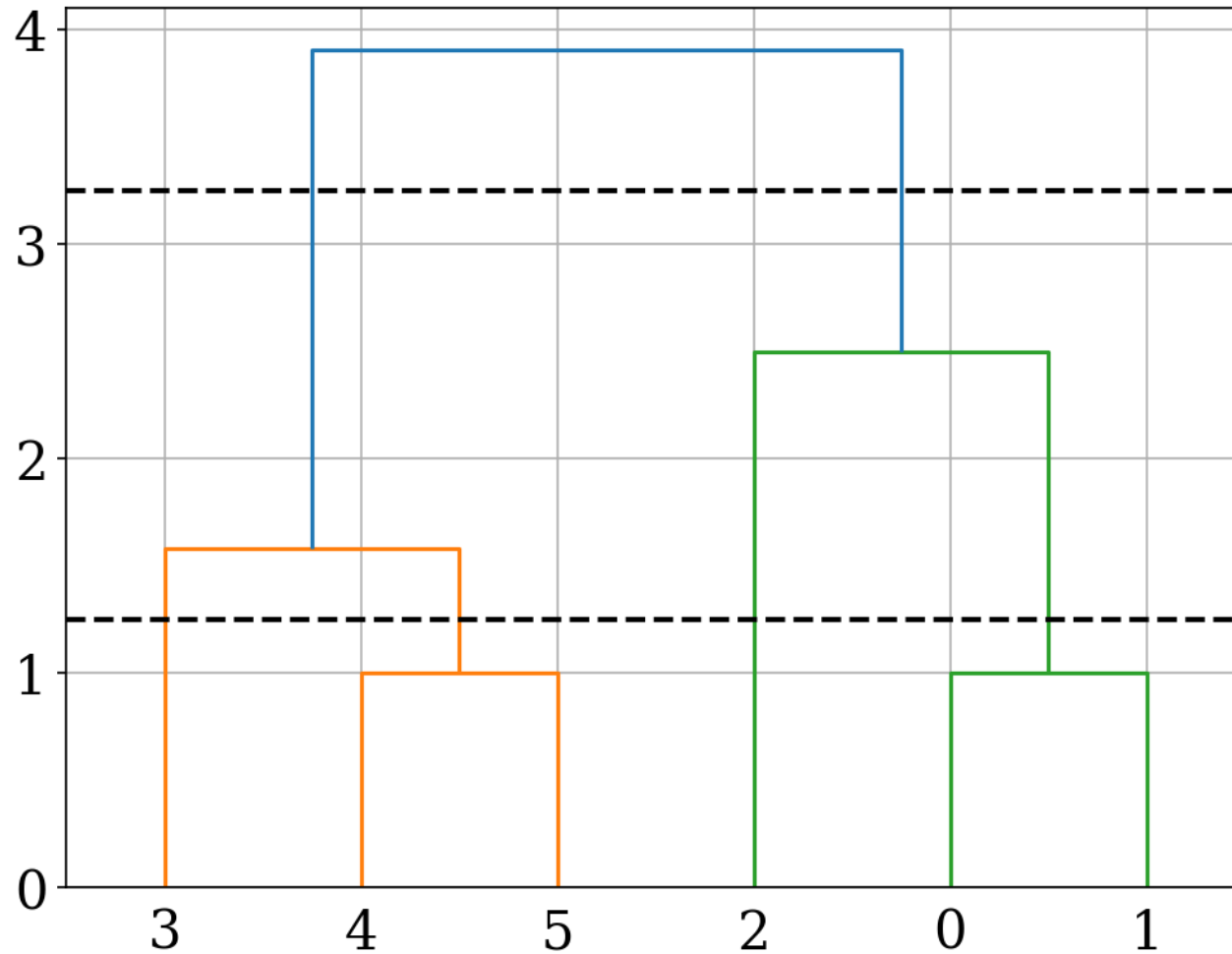
- Suppose two clusters  $C_i$  and  $C_j$  are two disjoint, non-empty sets of data.
- Agglomerative hierarchical clustering is based on a **pairwise dissimilarity metric**  $d(C_i, C_j)$ .
- The algorithm starts with  $n$  single membered clusters corresponding to each data point in the sample  $S_n$ .
- At each step, the algorithm merges two clusters that are most similar into a new cluster. The two clusters that minimize  $d(C_i, C_j)$  is merged together.
- The process is repeated until there is a single cluster of all data, after  $n - 1$  steps.

# Hierarchical Clustering

- The result is presented as a **dendrogram**, a tree where:
  - Each node represents a merge,
  - Leaf nodes are individual data points,
  - Root node contains all data.
- Typical dendrogram has the height of each node representing the dissimilarity between the children clusters.
- The **cophenetic distance** is the height of the lowest common parent node of two data points.
- Cutting the dendrogram at a given height produces a traditional assignment of points to clusters.
- Cutting at different heights produce different, nested clusterings.



# Hierarchical Clustering



# Hierarchical Clustering

- Different dendrograms are created by different pairwise cluster dissimilarity metrics.
- If  $d(\mathbf{X}, \mathbf{X}')$  is a distance metric like the Euclidean distance or 1-correlation between  $\mathbf{X}, \mathbf{X}' \in R^d$ .
- The most common pairwise dissimilarity metrics used are:

- Single-linkage dissimilarity

$$d_s(C_i, C_j) = \min\{d(\mathbf{X}, \mathbf{X}') \mid \mathbf{X} \in C_i, \mathbf{X}' \in C_j\}$$

- Complete-linkage dissimilarity

$$d_s(C_i, C_j) = \max\{d(\mathbf{X}, \mathbf{X}') \mid \mathbf{X} \in C_i, \mathbf{X}' \in C_j\}$$

- Average-linkage dissimilarity

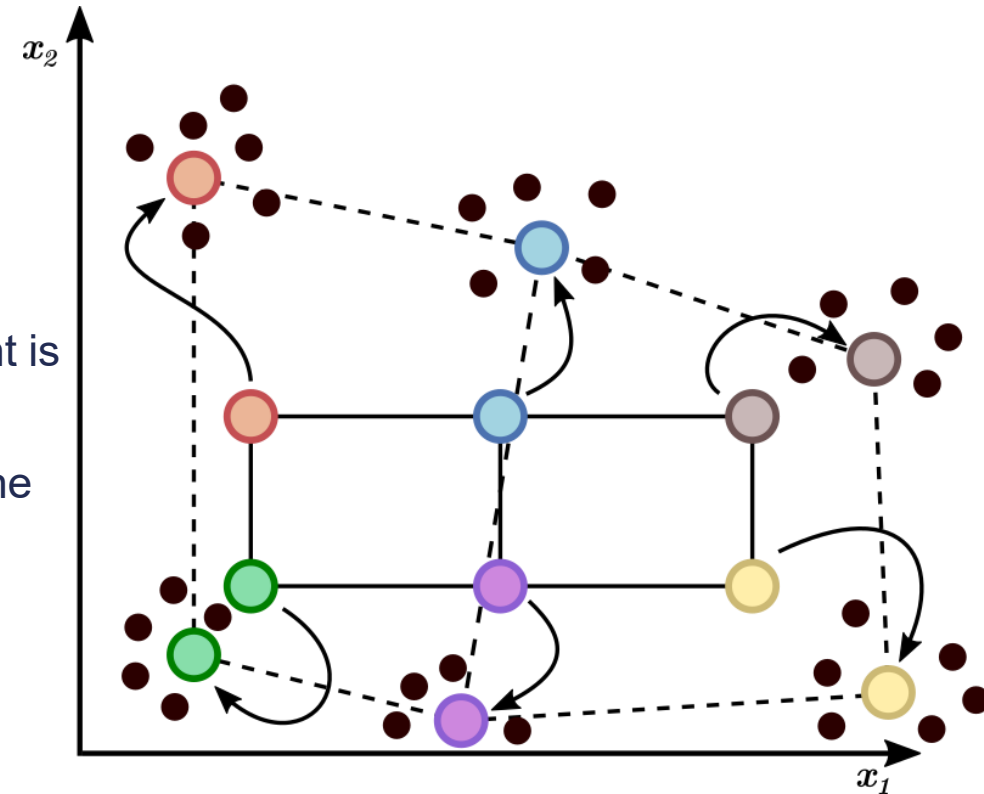
$$d_s(C_i, C_j) = \frac{1}{|C_i||C_j|} \sum_{\mathbf{X} \in C_i} \sum_{\mathbf{X}' \in C_j} d(\mathbf{X}, \mathbf{X}')$$

# Hierarchical Clustering

- Single-linkage is a type of nearest-neighbour metric.
- Complete-linkage is a maximum-distance criterion.
- Average linkage has an intermediate behaviour between these two extremes.
- Remember, that two data points that are close together in dendrogram doesn't mean they are close together in the original feature space.

# Self-Organizing Maps (SOM)

- Self-organizing maps iteratively adapts a grid of nodes to the data.
- The grid is typically two-dimensional and rectangular, but not necessarily.
- To begin with, the nodes of the grid are arbitrarily positioned in the feature space.
- A data point is selected at random, the grid node closest to that point is moved a certain distance towards it.
- Neighbouring nodes in the grid are made to move slightly towards the selected point.
- The process is repeated for a large number of iterations (20,000 to 50,000) until the grid adapts itself to the data.
- The data point closest to each node defines the clusters.
- The achieved clustering maintains the general structure of the initial grid.



# Self-Organizing Maps (SOM)

- Set up a grid of nodes, denoted by  $\mathbf{u}_1, \dots, \mathbf{u}_N$ .
- Let  $D(\mathbf{u}_i, \mathbf{u}_j)$  be the distances between any two nodes  $\mathbf{u}_i$  and  $\mathbf{u}_j$ .
- Initialize weight vectors  $f_0(\mathbf{u}_i)$  with same dimensionality as input data. They are initialized either randomly or using a heuristic.
- Randomly choose a datapoint,  $\mathbf{X}^{(k)}$ , which serves as the input for the current iteration  $k$ .
- Identify  $\mathbf{u}^{(k)}$ , the grid node whose weight vector  $f_k(\mathbf{u}^{(k)})$  is closest to  $\mathbf{X}^{(k)}$  such that  $\|f_k(\mathbf{u}^{(k)}) - \mathbf{X}^{(k)}\|$  is minimum, this is known as the **Best Matching Unit (BMU)**.



# Self-Organizing Maps (SOM)

- Update the weight vectors as:

$$f_{k+1}(\mathbf{u}_i) = f_k(\mathbf{u}_i) + \lambda(D(\mathbf{u}_i, \mathbf{u}^{(k)}), k) \cdot (f_k(\mathbf{u}_i) - \mathbf{X}^{(k)}), \quad \forall i = 1, \dots, N$$

where the learning rate  $\lambda$  is a decreasing function of both parameters  $D(\mathbf{u}_i, \mathbf{u}^{(k)})$  (distance between  $\mathbf{u}_i$  and  $\mathbf{u}^{(k)}$  **on the grid**) and  $k$ .

- $\lambda$  decreases as  $D(\mathbf{u}_i, \mathbf{u}^{(k)})$  increases, meaning nodes closer to the BMU on the grid are updated more than those farther away.
- $\lambda$  also decreases with  $(k)$ , reducing the overall update magnitude as training progresses.
- Often,  $\lambda(D, k) = \alpha(k) \cdot h(D)$ , where:
  - $\alpha(k)$  is a learning rate that decreases over time
  - $h(D)$  is a neighborhood function also decreasing over time.
- This update moves each node's weight vector toward  $\mathbf{X}^{(k)}$ , with the BMU moving the most and its neighbors moving less, proportional to their grid distance.

# Self-Organizing Maps (SOM)

- The process is repeated for a predefined number of iterations ( $K$ ) and each node  $\mathbf{u}_i$  with its final weight vector  $f_K(\mathbf{u}_N)$  acts as cluster center.
- For each new data point assign it to the cluster where the distance between the point and the weight vector is least.
- The SOM can be seen as a neural network:
  - The grid nodes  $\mathbf{u}_1, \dots, \mathbf{u}_N$  are the neurons,
  - The coordinates of the mapped points,  $f_k(\mathbf{u}_1), \dots, f_k(\mathbf{u}_N)$  corresponds to the weights, and
  - Each neuron is constrained to respond similarly to its neighbour.
- The process of adjusting the mapped points is similar to network training.
- This neural network is called a **Kohonen Network**.

# Thank You